Zhiyong Wang

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SUMMARY

AI Engineer with 1+ year shipping agentic AI in production — MCP clients and servers, tool-calling agents with memory systems, and multi-step workflow automation on Amazon Bedrock and LangChain/LangGraph. 4+ years of production ML (SageMaker, PyTorch, feature stores, Airflow, MLflow) and 10+ years of software engineering across big data, distributed systems, and cloud. Seeking Agentic AI Engineer roles.

TECHNICAL SKILLS

Languages: Python, C/C++; familiar with Java, Scala

AI & LLM: LangChain/LangGraph, Amazon Bedrock, MCP (client & server), AI agents (tool calling, memory systems), prompt-driven workflows, Featurestores(Tecton)

Cloud & Infra: AWS (S3, EMR, Lambda, API Gateway, DynamoDB, ECS, Athena, OpenSearch), Linux

ML & DL: PyTorch, scikit-learn, LightGBM/XGBoost, MLflow, AWS SageMaker

Data Engineering: Apache Spark, Kafka, Databricks, Airflow, Snowflake, SQL (MySQL, PostgreSQL), Redis

Tools: Git, Jupyter, Anaconda, Streamlit

EXPERIENCE

Machine Learning Engineer (Full-time) — Tempo.io ***Oct 2021 – Sep 2026***

- Built MCP clients and servers enabling AI agents to query Snowflake through structured tool calls, implementing robust tool interfaces, error handling, and reliable multi-step execution.
- Developed AI agents integrating multiple MCP servers, external tools, and persistent memory to automate time management, task prioritization, and complex user workflows.
- Designed and implemented a RAG pipeline using Amazon OpenSearch as a vector database, integrating document ingestion, embedding generation, semantic retrieval, and contextual grounding to improve agent response accuracy.
- Deployed containerized AI agents and MCP servers on Amazon EKS (Kubernetes), implementing horizontal pod autoscaling, health probes, rolling deployments, and centralized logging to improve service scalability, reliability, and operational visibility.
- Developed an automated AI agent evaluation harness combining LLM-as-a-Judge, deterministic validation, and execution trace analysis to assess retrieval relevance, tool-calling accuracy, Snowflake query correctness, response groundedness, and end-to-end task completion.
- Designed and built an automatic time-management system collecting user activity signals from Jira, Google Calendar, and VS Code/IDEs, rendering ranked task suggestions and time-awareness insights.
- Fine-tuned a sentence-transformer (bi-encoder) model on Jira ticket data from the Atlassian Jira system and Tempo's Jira data for semantic similarity search — 5.3% improvement across all metrics over the baseline NLU-based ticket search.
- Built, trained, and deployed learning-to-rank and time-awareness models with SageMaker, scikit-learn, and PyTorch — Top-1 accuracy 75%, Top-5 accuracy 91.5% and NDCG 0.86.
- Built evaluation pipelines with labeled Jira datasets and tracked Top-1/Top-5 accuracy and NDCG (0.86), running regression checks on every model iteration before each production release.
- Designed feature-engineering pipelines and implemented them in a feature store(Tecton) serving both training and inference.

- Built and deployed a real-time analytics dashboard on AWS (API Gateway, Lambda, DynamoDB, OpenSearch) for user activity data.
- Productionized models through automated training-and-deployment pipelines (Airflow orchestration, SageMaker) with production monitoring dashboards on AWS (API Gateway, Lambda, DynamoDB, OpenSearch).

Machine Learning Engineer (Contract) — Expedia ***Aug 2021 – Oct 2021***

- Built Spark data pipelines on Databricks, orchestrated with Airflow, feeding a feature store for ML models.
- Developed ML models (SageMaker, scikit-learn, PyTorch) predicting click-through and booking rates for travel packages (flights, hotels, car rentals).
- Deployed a Streamlit dashboard on AWS for real-time airplane and hotel booking analytics.

Staff Software Engineer — Calix Inc. ***Oct 2012 – Jul 2019***

- Designed and developed a big-data platform for smart-home systems: real-time and batch processing with Spark SQL, DataFrames, and Structured Streaming.
- Built ML models with Spark MLlib, scikit-learn, and PyTorch (regression, classification, clustering, LightGBM) and evaluated them at scale.
- Owned microservices end to end; built RESTful APIs with Java/Spring; optimized MySQL/Redis schemas and queries.

Senior Software Engineer — ZTE Corp ***Aug 2004 – Aug 2012***

- Designed and developed distributed in-memory databases and fault-tolerant telecom systems in C/C++; debugged complex systems with the GNU toolchain (gdb, gprof).

SELECTED PROJECTS

- Agentic AI platform (Tempo): Built MCP clients and servers enabling AI agents to query Snowflake through structured tool calls, with persistent memory across sessions; deployed containerized on Amazon EKS with horizontal pod autoscaling, health probes, and centralized logging.
- Agent RAG & evaluation harness (Tempo): Designed a RAG pipeline on Amazon OpenSearch (document ingestion, embedding generation, semantic retrieval, contextual grounding) and an automated eval harness (LLM-as-a-Judge, deterministic validation, trace analysis) assessing retrieval relevance, tool-calling accuracy, and end-to-end task completion.
- Fine-tuned a sentence-transformer (bi-encoder) model on Jira ticket data for semantic similarity search — 5.3% improvement across all metrics over the baseline NLU-based ticket search.

EDUCATION

- M.S., Data Science — University of British Columbia
- M.S., Electrical & Computer Engineering — University of Massachusetts Amherst
- B.S., Electric Information Science and Technology — Nanjing University of Posts and Telecommunications